Isa (Berk) Geriler

Graphics / Rendering Programmer · MSc Games Engineering, University of Warwick
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EDUCATION

MSc Games Engineering, University of Warwick (WMG) *Sep 2025 – Sep 2026*

- Dissertation: *Improving Sampling for Real-time Rendering (Path Guiding)*, supervised by Dr. Thomas Bashford-Rogers.
- Modules: Computer Graphics, Advanced Computer Graphics, Games Engine Design & Development, Games Engineering, Programming & Fundamental Algorithms, Innovative Simulation Design and Development, Fundamentals of Games Research & Management.

BSc Software Engineering, Isik University *Aug 2020 – Jul 2025*

- Grade: **3.29 / 4.0 GPA (UK Classification: 2:1).**
- Relevant coursework: Calculus II, Calculus III, Linear Algebra, Data Structures & Algorithms, Analysis of Algorithms, Deep Learning in Artificial Neural Networks, Introduction to Computer Vision, Computer Organization, Operating Systems, Software Architecture, Human-Computer Interaction.

PROJECTS

Path Tracer & Light Transport Engine (C++)
University of Warwick

- Extended, CPU-based, multithreaded physically based renderer built to explore light transport and microfacet models.
- Supports three integrators in a single architecture: Path Tracing, Instant Radiosity, and Light Tracing. Instant Radiosity uses a Halton quasi-Monte Carlo sampler relying on prime bases for the Radical Inverse.
- Material framework covers GGX microfacets (Conductor BSDF, sampled proportionally to the NDF), Plastic BSDF (Phong model), Oren-Nayar BSDF, and a Layered BSDF that evaluates Beer's Law for attenuation.
- Manages render time and variance with a Binned SAH BVH, tile-based rendering, and Multiple Importance Sampling for latitude-longitude environment maps using a luminance-based PDF.
- Outputs custom AOVs to feed Intel Open Image Denoise (OIDN) for post-process denoising.

Software Rasterizer (C++, SIMD)
University of Warwick

- Built a 3D rendering engine from scratch that simulates the GPU graphics pipeline on the CPU. A minimal backend (GamesEngineeringBase) handles only window management and pixel plotting; all rendering logic and mathematics are implemented by hand.
- Triangle rasterization using edge functions and barycentric coordinates.
- Custom 3D math library written from first principles: 4×4 matrix transforms, vector operations, quaternions, and homogeneous coordinates.
- Full Model-View-Projection pipeline, including a derived camera LookAt matrix and perspective projection matrix.
- Perspective-correct attribute interpolation for vertex normals and colours (1/w depth correction) to remove affine texture-mapping artefacts.
- Depth buffer (Z-buffer) for visibility testing and implicit backface culling via edge-function winding order.
- Custom shader model with Lambertian diffuse and ambient lighting from interpolated vertex normals; parses .gem meshes (geometry and skeletal data) via GEMLoader.
- Optimized with SIMD (SSE/AVX/AVX2) and multithreading.

Offline Ray Tracer (C++, in progress)

- Ongoing multithreaded CPU ray tracer used as a testbed for light-transport math.
- Implements the full Ray Tracing in One Weekend (Shirley et al., 2025) architecture and extends into book two with motion blur and a custom Bounding Volume Hierarchy (BVH) to reduce spatial-intersection cost.
- BVH build parallelized with C++17 std::execution::par to keep the CPU saturated; continuing through the rest of the series.

DX12 Engine (DirectX 12, in progress)

- Custom rendering engine built from scratch in DirectX 12 to work at the API level.
- Focus on low-level concepts: descriptor heaps, root signatures, and pipeline state objects (PSOs).

- A re-architected, improved version of an earlier coursework framework.

Skyfire Uprising, Level 2 (Unreal Engine 5, Team Project)
University of Warwick
- Team game built in Unreal Engine 5; I owned the visual look of the game.
- Authored custom materials and shaders (such as Gerstner Wave Water Shader), post-process effects, volumetric fog, and lighting.

Real-Time Face Recognition System for Authentication (Python)
Işık University · Team graduation project
- On-premise face-recognition system to replace ID-card and QR-code authentication at the university, built to comply with data-privacy regulation (KVKK), processing biometric data in a controlled Google Colab environment.
- Pipeline detects live faces in a video stream with OpenCV's Haar Cascade classifier, then generates a 512-dimensional embedding per face with a pre-trained FaceNet model.
- Stores embeddings in PostgreSQL with the pgvector extension and runs L2 / Euclidean KNN similarity search, verifying identities in under 5 seconds.
- Flask web dashboard for staff with role-based access control, two-factor authentication (pyotp), and real-time event logging.
- **Selected to present the project poster at the 33rd IEEE SIU 2025 Conference** (Signal Processing and Communications Applications), Undergraduate Projects Competition.

Chat Room (C++, Dear ImGui, FMOD)
- Client-server chat application built from scratch using WinSock; the server handles multiple clients concurrently.
- Client GUI built with Dear ImGui, supporting public broadcast messages and private 1-to-1 direct messages.
- FMOD integration for real-time sound notifications on incoming messages.

EXPERIENCE

Student Assistant, Işık University — *Apr 2024 – May 2025*
Şile, Istanbul · On-site
- Assisted over 100 students in programming and microprocessor labs covering Java, object-oriented programming, data structures, and Arduino/C++.
- Managed grading, quiz evaluation, and exam proctoring across multiple CSE/SWE courses.

Backend Developer Intern, T-Rupt — *Jul 2024 (1 Month)*
Maslak, Istanbul · On-site
- Developed an Excel importer/exporter in Java (Maven) to automate data transfer.
- Built a RESTful API in Spring Boot with Docker containerization, and authored OpenAPI v3.0 documentation to support integration.

Software Engineer Intern, Ordinatrum — *Aug 2023 – Sep 2023*
Kadıköy, Istanbul · Remote
- Worked on automation and machine-learning solutions in a healthcare-informatics setting.
- Built and queried databases in Microsoft SQL Server, strengthening SQL and data-handling skills.

SKILLS

Languages: C++, Python, HLSL, Java, SQL
Graphics & Rendering: DirectX 12, Ray Tracing / Path Tracing, BVH Acceleration, Multiple Importance Sampling, GGX / Microfacet BSDFs, SIMD (SSE/AVX/AVX2), Multithreading, RenderDoc
Engines & Tools: Unreal Engine 5 (materials, shaders, lighting), Dear ImGui, FMOD, WinSock, OpenCV, FaceNet, PostgreSQL / pgvector, Flask, Spring Boot, Docker, REST / OpenAPI, Git
Spoken languages: Turkish (native), English (IELTS 8.0, C1), Japanese (beginner)

CERTIFICATIONS

- **IELTS Academic**, Overall Band 8.0 (CEFR C1), British Council. Issued Feb 2025 (valid to Feb 2027).

VOLUNTEERING

Management Team Member, Isik Siber (University Cyber-Security Society) — *Jun 2022 – Feb 2023*
- Helped align strategy and operations for the university cyber-security society.
- Organized events to support collaborative learning and networking.
- Encouraged knowledge sharing and collaboration among members.